

POWERFACTORY

PowerFactory 2024

Technical Reference

DIgSILENT Variable Speed Drive Template

DIgSILENT VSD 200kW

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 Introduction

This document describes the Variable Speed Drive (VSD) template “VSD_200kW” that is delivered with *PowerFactory*, its applicability and possible configurations.

The “VSD_200kW” template is a generic model of a variable speed drive based on a diode bridge rectifier i.e. passive front-end (on the supply side) and a two level voltage source inverter (on the motor side). An intermediate circuit composed of a positive and a negative DC bus along with a DC capacitor interfaces the rectifier with the inverter. The AC/DC rectifier is rated to 0.56 kV AC while a transformer is included in the template to provide a standard 0.4 kV AC supply terminal. If necessary, the transformer voltage ratings can be freely adjusted in order to meet the voltage ratings of the connection point.

The template can be used for the following studies:

- Balanced/unbalanced AC load flow analysis
- Steady state speed control in RMS- or EMT-Simulation
- Motor starting in RMS- or EMT-Simulation
- Fault ride through studies including motor re-acceleration function in RMS- or EMT-Simulation

2 Model Description

The VSD model is represented in the single line diagram by a rectifier (*ElmRec*), a DC capacitor (*ElmScap*) and a PWM converter (*ElmVsc*) connected to an induction motor (*ElmAsm*), as shown in Figure 2.1.

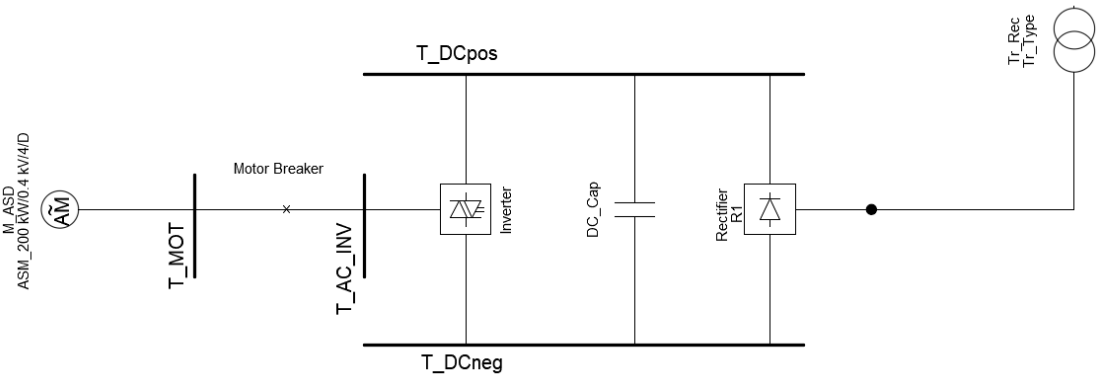


Figure 2.1: Generic *Variable Speed Drive* - Single Line Diagram

The overview of the VSD dynamic control model is represented using the composite model *VSD_Model*, as shown in Figure 2.2. This composite model can be found either through the link *Model* on the Basic Data page of the PWM Converter or using the Data Manager and locating it within the corresponding grid folder where the template has been deployed.

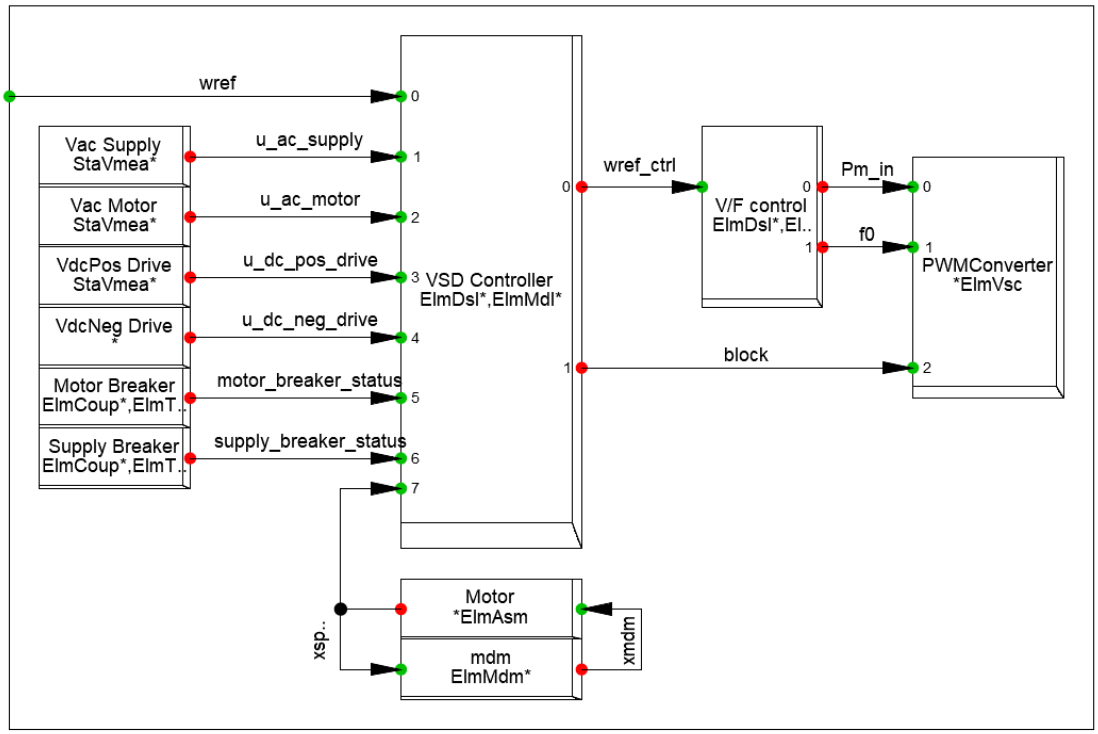


Figure 2.2: Generic *Variable Speed Drive* Controller Overview

The main components of this model are:

- Control model slots for measurements and breaker status flag;
- Motor slot for sensing actual speed
- Motor driven machine slot for coupling with the motor mechanical model. A load characteristic can be programmed within this model.
- PWM Converter slot for sending control references to the motor side inverter
- VSD Controller slot for linking the main VSD controller with measurements and the inverter
- V/F control block for including a generic constant V/f control law

The shown slots are briefly described in Table 2.1. Each slot is assigned to either a DSL model or a *PowerFactory* element, such as a asynchronous machine (*ElmAsm*).

Table 2.1: Description of Composite Frame Slots

Slot Name	Description	Needed Type
Motor	Slot for the driven Induction Motor (set as Main slot - if the motor is set to “out of service” then the composite model (and all its elements within it) will be disabled)	ElmAsm
mdm	Slot for the Motor Driven Machine element	ElmMdm
Vac Supply	AC supply side voltage measurement	StaVmea
Vac Motor	AC motor side voltage measurement	StaVmea
VdcPos	DC side voltage measurement (positive rail)	StaVmea
VdcNeg	DC side voltage measurement (negative rail)	StaVmea
Motor Breaker	Slot keeping a reference to the motor breaker	ElmCoup,ElmTerm,StaSwitch
Supply Breaker	Slot keeping a reference to the AC supply breaker	ElmCoup,ElmTerm,StaSwitch
VSD Controller	Slot for the dynamic VSD controller	ElmDsl,ElmMdl
V/f control	Slot for look up table of the V/f references	ElmDsl,ElmMdl
PWM Converter	Slot for the PWM Converter (motor side inverter unit) (set as Main slot - if the motor is set to “out of service” then the composite model (and all its elements within it) will be disabled)	ElmVsc

The VSD dynamic control model is implemented using DSL, within the block definition “VSD Controller”, as shown in Figure 2.3.

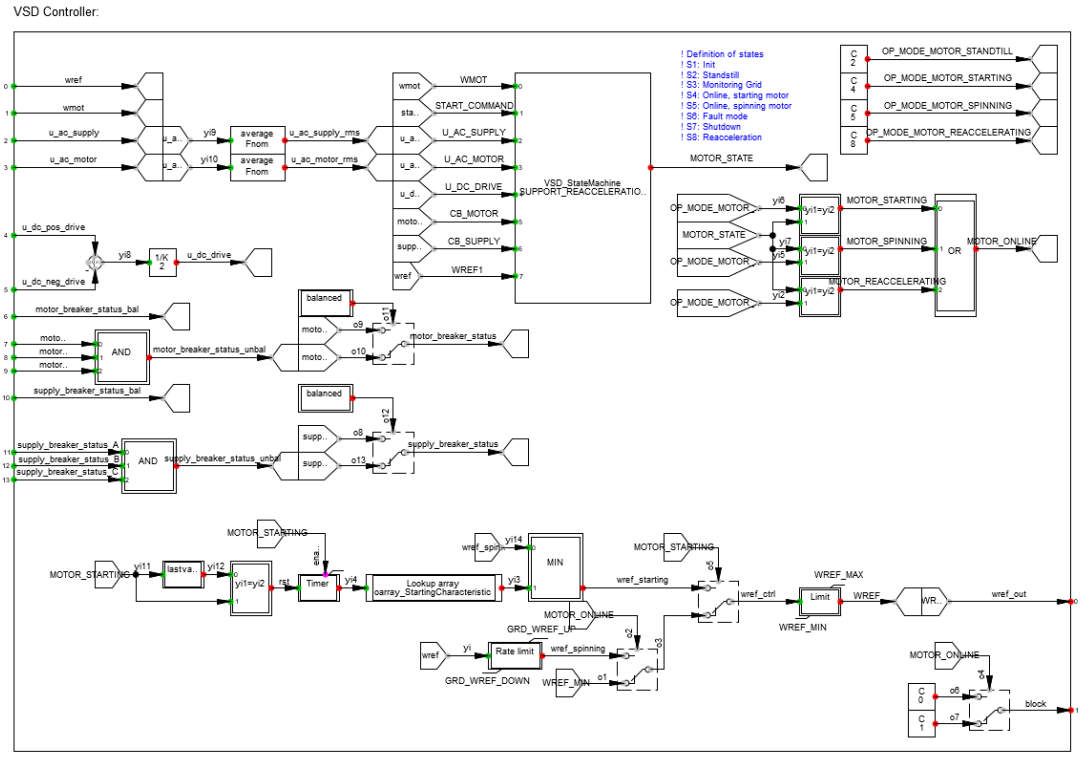


Figure 2.3: Generic *Variable Speed Drive* Controller Implementation

The main components/functions of the control algorithm are:

- Average value calculation of AC voltage inputs
- Control state machine
- Speed reference limiters
- Speed versus time characteristic for motor starting (*StartingCharacteristic*)

The dynamic controller parameters are described in Table 2.2.

Table 2.2: VSD Controller model parameters

Parameter Name	**Unit**	**Description**
SUPPORT_REACCELERATION	0/1	Support re-acceleration function
RECONNECT_COUNTER_MAX	-	Maximum number of motor starting re-trials before shutting down
U_MIN_CONNECT	p.u.	Minimum supply voltage at which motor may connect
U_MAX_CONNECT	p.u.	Maximum supply voltage at which motor may connect
U_MIN_SUPPLY	p.u.	Minimum allowed supply voltage during normal operation
U_MAX_SUPPLY	p.u.	Maximum allowed supply voltage during normal operation
U_MAX_MOTOR	p.u.	Maximum allowed motor voltage during normal operation
W_STARTED_THRESHOLD	p.u.	Threshold for motor being considered as started at initialisation
W_BAND_STEADYSTATE	p.u.	Allowed band around speed reference
T_DETECT_STALLING	s	Stalling detection time
T_U_RECONNECT	s	Reconnect time after a fault (voltage monitoring)
T_W_RECONNECT	s	Reconnect time after a fault (speed monitoring)
T_MIN_W	s	Minimum time of steady state speed for entering normal operation
T_W_REACCELERATE	s	Minimum wait time before trying to re-accelerate
T_FAULT_U_SUPPLY	s	Maximum time at which supply voltage can exceed limits
T_FAULT_U_MOTOR	s	Maximum time at which motor voltage can exceed upper limit
T_FAULT_MAX_ALLOWED	s	Maximum fault period before shutting down
Fnom	Hz	Nominal grid frequency
WREF_MIN	p.u.	Minimum reference speed
GRD_WREF_DOWN	p.u./s	Speed reference gradient limiter (downwards)
WREF_MAX	p.u.	Maximum reference speed
GRD_WREF_UP	p.u./s	Speed reference gradient limiter (upwards)

The core part of the VSD Controller is the VSD State Machine, with the logic described graphically within Figure 2.4.

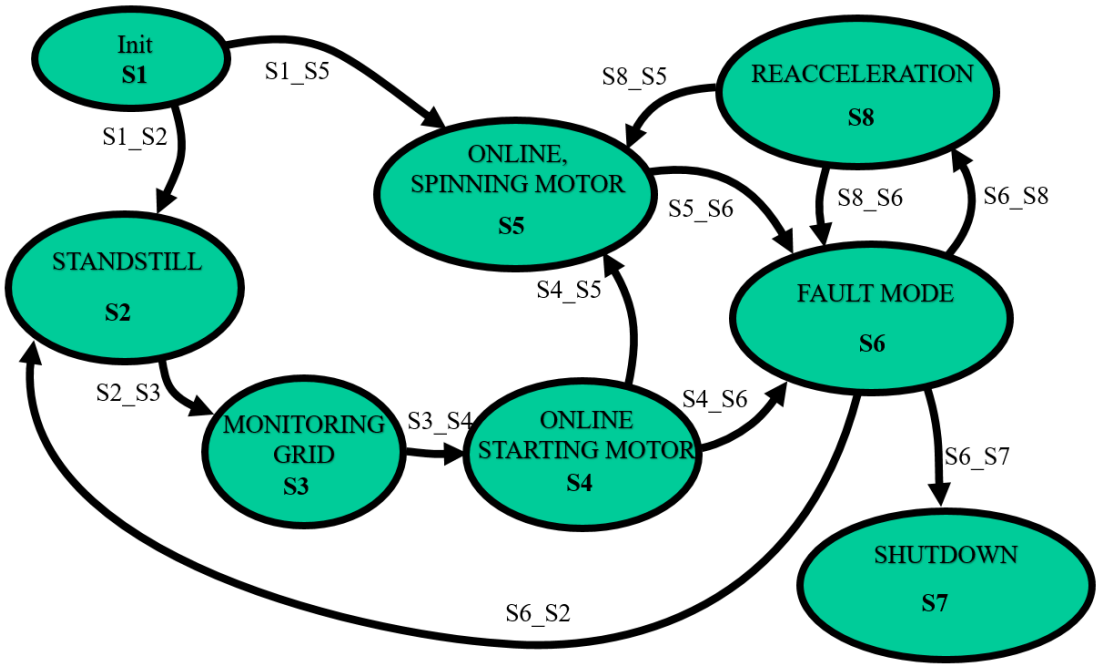


Figure 2.4: Generic *Variable Speed Drive* Controller State Machine

The states in which the VSD can exist are listed below:

- S1 - Motor initialisation state. The motor exists in this state only at initialisation. A decision is made automatically whether the load flow condition corresponds to a running or a standstill operation scenario.
- S2 - Motor standstill state. In this state, the motor is at standstill, the drive is not energising the motor, only awaits for a motor start command.
- S3 - Monitoring grid state. Upon triggering the motor start command (manually done by the user, via a parameter event), the VSD monitors the AC side supply voltage. If the AC grid conditions are within acceptable range, the motor starting is triggered automatically.
- S4 - Online, motor starting state. In this state, the motor is ramped up to reference speed. The reference speed can be non-nominal as well. The VSD can trigger a fault if certain conditions exist (for example if the motor is stalling) or it will automatically transfer to the spinning motor state if a successful motor start is detected.
- S5 - Online, spinning motor state. This state is the normal operation mode when the motor is running at reference speed. The VSD can transfer into a fault mode state if certain conditions are fulfilled.
- S6 - Fault mode state. This state can occur whenever the motor behaviour is outside its expected specification or in case of out of range parameters of the AC side supply. Depending on the VSD settings, the VSD will try to exit this state by either reconnection (manual, via parameter) or reacceleration (automatic) of the motor.
- S7 - Shutdown state. After a specific number of fault mode state re-entries, the VSD will permanently enter the shutdown state in which the AC supply side and motor side breakers are commanded to open. No further reconnections are possible
- S8 - Reacceleration state. This state is intended for automatic motor re-acceleration after an AC side fault event e.g. low voltage fault event. The motor is re-accelerated up to

reference speed during this stage via predefined ramps. The motor starting (speed-vs-time) characteristic is not taken into account in this state. Upon successful re-acceleration, the motor re-enters the online, spinning motor state. If unsuccessful, the VSD re-enters into fault mode state.

The state transitions (transition of state machine from state Sx to state Sy) are defined as "Sx_Sy" further below:

- **S1_S2**: "Init" to "STANDSTILL". This transition occurs if:

$$\omega_{mot} < \text{W_STARTED_THRESHOLD}$$

- **S1_S5**: "Init" to "ONLINE, SPINNING MOTOR". This transition occurs if:

$$\omega_{mot} >= \text{W_STARTED_THRESHOLD}$$

- **S2_S3**: "STANDSTILL" to "MONITORING GRID". This transition occurs if:

$$start_command > 0.5$$

  The post-transition actions are:

  - close Motor breaker (if not already closed);
  - $start_command := 0$.

- **S3_S4**: "MONITORING GRID" to "ONLINE, STARTING MOTOR". This transition occurs if:

$$\begin{aligned} & \text{Motor breaker is closed} \\ and \quad & \text{U_AC_SUPPLY} > \text{U_MIN_CONNECT} \\ and \quad & \text{U_AC_SUPPLY} < \text{U_MAX_CONNECT} \end{aligned}$$

  The post-transition actions are:

  - close Supply breaker (if not already closed)

- **S4_S5**: "ONLINE, STARTING MOTOR" to "ONLINE, SPINNING MOTOR". This transition occurs if:

$$abs(\omega_{mot} - \omega_{ref}) < \text{W_BAND_START}$$

- **S4_S6**: "ONLINE, STARTING MOTOR" to "FAULT MODE". This transition occurs if:

$$\begin{aligned} & abs(\omega_{mot} - \omega_{ref}) > \text{W_BAND_STEADYSTATE}\ for\ \text{T_DETECT_STALLING}\ s \\ or \quad & \text{Motor breaker open} \\ or \quad & \text{Supply breaker open} \end{aligned}$$

- **S5_S6**: "ONLINE, SPINNING MOTOR" to "FAULT MODE". This transition occurs if:

$$\begin{aligned} & \text{U_AC_SUPPLY} < \text{U_MIN_SUPPLY}\ for\ \text{T_U_FAULT}\ s \\ or \quad & \text{U_AC_SUPPLY} > \text{U_MAX_SUPPLY}\ for\ \text{T_U_FAULT}\ s \\ or \quad & \text{U_AC_MOTOR} > \text{U_MAX_MOTOR}\ for\ \text{T_U_FAULT}\ s \\ or \quad & abs(\omega_{mot} - \omega_{ref}) > \text{W_BAND_STEADYSTATE}\ for\ \text{T_DETECT_STALLING}\ s \\ or \quad & \text{Motor breaker open} \\ or \quad & \text{Supply breaker open} \end{aligned}$$

- **S6_S2**: “FAULT MODE” to “STANDSTILL”.This transition occurs if:

$$(\text{U_AC_SUPPLY} > \text{U_MIN_SUPPLY}\, and\, \text{U_AC_SUPPLY} < \text{U_MAX_SUPPLY})\, for\, \text{T_U_RECONNECT}\, s$$
$$and \quad \omega_{mot} < \text{W_MIN_REF}\, for\, \text{T_W_RECONNECT}\, s$$

  The post-transition actions are:

  – $RECONNECT_COUNTER+ = 1$

- **S6_S7**: “FAULT MODE” to “SHUTDOWN”. This transition occurs if:

$$\text{RECONNECT_COUNTER} > \text{RECONNECT_COUNTER_MAX}$$
$$or \quad \text{time spent once in FAULT MODE state} > \text{T_FAULT_MAX_ALLOWED}$$

  The post-transition actions are:

  – Motor Breaker open
  – Supply Breaker open

- **S6_S8**: “FAULT MODE” to “REACCELERATION”. This transition occurs if:

$$\text{SUPPORT_REACCELERATION} = 1$$
$$and\, (\text{U_AC_SUPPLY} > \text{U_MIN_SUPPLY}\, and\, \text{U_AC_SUPPLY} < \text{U_MAX_SUPPLY})\, for\, \text{T_U_REACCELERATE}\, s$$

- **S8_S5**: “REACCELERATION” to “ONLINE, SPINNING MOTOR”. This transition occurs if:

$$abs(\omega_{mot} - \omega_{ref}) < \text{W_BAND_START}$$

- **S8_S6**: “REACCELERATION” to “FAULT MODE”. This transition occurs if:

$$\text{U_AC_SUPPLY} < \text{U_MIN_SUPPLY}\, for\, \text{T_U_FAULT}\, s$$
$$or \quad \text{U_AC_SUPPLY} > \text{U_MAX_SUPPLY}\, for\, \text{T_U_FAULT}\, s$$
$$or \quad abs(\omega_{mot} - \omega_{ref}) > \text{W_BAND_STEADYSTATE}\, for\, \text{T_DETECT_STALLING}\, s$$
$$or \quad \text{Motor breaker open}$$
$$or \quad \text{Supply breaker open}$$

3 How to use this Template

PowerFactory contains pre-defined templates of dynamic models for different generic power system equipment types. The global library template “VSD_200kW_0.4kV” is located under *DIgSILENT Library* → *Templates*→ *Variable Speed Drives*.

Using the template is straightfoward:

- click the **General Templates** button ( ) from the Drawing Toolbar
- select (left click once) the VSD template in the subsequent dialog.
- position the mouse in the single line diagram and left click once: the undeployed VSD diagram is shown.

- Move the mouse towards an area in the single line diagram that offers enough empty space to contain the VSD and then left click once more to deploy it.
- The VSD is now deployed in the system (including all dynamic controllers) and can be connected to the network
- Right click the VSD block and choose “Show Graphic” to display the internal representation of the VSD including the transformer, motor, inverter and rectifier elements.
- The VSD comes with an AC side “Supply Breaker” element connected to a terminal “T_AC_Motor”. Do not remove these elements from the model as they are referenced to the dynamic control model. If removed, the dynamic model may not operate properly or even generate an error.
- connect/use the terminal “T_AC_Motor” further and connect it to the AC system.
- If the AC side voltage requires adaption to a different rating, then do the following:
  - Modify the rated voltage of the two terminals to which the “Supply Breaker” is connected e.g. “T_AC_Motor” and “T_AC_Motor_internal”.
  - open the detailed VSD representation (*right click on template* → *Show Graphic*) and open the transformer type (*TypTr2*) dialog. Modify the rated voltage of the transformer accordingly.

Note: by default, the transformer HV side is the one towards the rectifier, and the LV side towards the AC supply network. If the new voltage rating is below 0.56kV, then the LV side of the transformer should be adapted, no other modifications needed. If the new voltage rating is above 0.56 kV e.g. 10kV, then the HV side rating must be set to e.g. 10 kV and the LV side to 0.56 kV. Subsequently, click on the “Flip Connections” in the transformer element (*ElmTr2*) dialog.

If you want to delete a model you have to keep in mind that the model consists of visible elements in the single line diagram and the additional composite model which is only accessible from the Data Manager (e.g. within the corresponding *Grid* folder of the project). Delete both the single line diagram elements and the composite model for a complete VSD model removal.

4 How to Customise the Model

You will find in this section a description on how you can configure the model from the template according to your specific needs. The configuration is sorted to the calculation functions because there are fewer configurations needed for load flow calculation than for a dynamic simulation.

4.1 Load Flow Analysis

The template supports in Load Flow calculation both operation modes (in service / not operating) as follows:

- both VSD and motor in service - the VSD “Supply breaker” is closed and the motor is operating (switch “Motor Breaker” is closed)
- VSD in service, motor disconnected - the VSD “Supply breaker” is closed and the motor is not operating (“Motor Breaker” is open)

- VSD and motor disconnected - the VSD “Supply breaker” is open and the motor is not operating (“Motor Breaker” is open)
- VSD disconnected (i.e. Supply breaker open), motor in service (i.e. Motor breaker closed) - this operation scenario is identical in behaviour with the previous case, whereas the load flow calculation will de-energise the motor.

4.1.1 Changing the rated voltage of the VSD’s AC supply side

If the AC supply side voltage requires adaption to a different voltage rating, then do the following:

- Modify the rated voltage of the two terminals to which the “Supply Breaker” is connected e.g. “T_AC_Motor” and “T_AC_Motor_internal”.
- open the detailed VSD representation (*right click on template* → *Show Graphic*) and open the transformer type (*TypTr2*) dialog. Modify the rated voltage of the transformer accordingly.

Note: by default, the transformer HV side is the one towards the rectifier, and the LV side towards the AC supply network. If the new voltage rating is below 0.56kV, then the LV side of the transformer should be adapted, no other modifications needed. If the new voltage rating is above 0.56 kV e.g. 10kV, then the HV side rating must be set to e.g. 10 kV and the LV side to 0.56 kV. Subsequently, click on the “Flip Connections” in the transformer element (*ElmTr2*) dialog.

4.1.2 Changing the rated voltage of the rectifier

Typically, the rectifier supply side voltage is not changed. The rated voltage of the AC supply side can be changed either by only varying the rated voltage on the grid side of transformer “Tr_Rec” (as indicated in the previous subsection), Should the rectifier side rated AC voltage be changed, then several related parameters must be adapted:

- Change the rated AC voltage of the rectifier type (Basic Data page of the object “Rectifier Type”)
- Configure accordingly the transformation ratio of the converter transformer “Tr_Rec”, i.e. the LV side rating of the transformer type needs to be identical to the rectifier rating, while the HV side rating needs to be identical to the AC supply voltage.
- Change the rated DC busbar voltage of the positive and negative DC terminals according to the calculated rating in the rectifier type (Basic Data page) - i.e. half the rated rectifier DC voltage on each terminal
- Change the rated DC voltage of the DC capacitor to a value equal to the rated rectifier DC voltage
- Change the rated DC voltage of the inverter element

4.1.3 Changing the rated power of the VSD

Rated power needs to be changed in several locations:

- Rated power of the rectifier type (in Basic Data page of the type). Set this value so that it accounts for the motor rated power plus the total losses in the VSD;
- Rated power of the supply side transformer type (it has typically a similar rating as the rectifier)
- Rated power of the PWM converter (in Basic Data page of the PWM converter element). Set this value so that it accounts for the motor rated power plus the losses in the cable (if existing) and converter
- Motor type data and load flow dispatch must be updated to fit the new power rating specifications

4.1.4 Changing the transformer type

The transformer type can be changed directly from the single line diagram, as it is typically done (e.g. open the *Edit* Dialogue of the transformer element and change the referenced Transformer type). Special care should be taken in the Rectifier element, where the commutation reactance should be updated according to the newly changed transformer reactance (i.e. compute the ohmic reactance of the transformer and set this value to parameter “Commutation Reactance” in the Load Flow page of the rectifier element - refer to below figure).

4.2 Short-Circuit Calculation (VDE / IEC / ANSI / complete)

The short-circuit behaviour of the model is completely determined by the elements interfacing with the AC side. The IEC 60909 standard mentions that a short-circuit current contribution shall be accounted for VSDs only in the case of reversible drives (i.e. bidirectional rectifiers). As the topology chosen in this template does not allow a power flow from the intermediate circuit (DC capacitor) to the supply side, the short-circuit current contribution is set to zero.

4.3 RMS- and EMT-Simulation

The main purpose of the model is to observe the behaviour of the VSD using a dynamic simulation.

An already prepared example is provided at the following link: https://www.digsilent.de/en/faq-reader-powerfactory/do-you-have-an-example-showing-how-to-model-a-variable-speed-driver.html which can be used to demonstrate any of the below documented functions.

The model is generic and open - you may configure/change it according to your needs. A short description of various useful parameterisation procedures is detailed in this section. For RMS simulations, the following settings are recommended:

- For fixed time step simulations: Simulation time step: 1 ms
- For variable step size simulations: minimum time step 1 ms, maximum time step 10 ms.

For EMT simulations, the following settings are recommended:

- For fixed time step simulations: Simulation time step: 0.1 ms
- For variable step size simulations: minimum time step 0.1 ms, maximum time step 8 ms.

4.3.1 Initial state of simulation

The motor may be configured to start initially either from a steady state point close to nominal speed or from a standstill point. This is achieved via the adjustment of the “Motor Breaker” and “Supply Breaker” based on the load flow configuration detailed in Section4.1. To start the simulation from motor standstill set one of the breakers to open (either motor side or supply side). To start the simulation from nominal speed, close both breakers, run a load flow and then start the simulation. It is possible to set a certain motor load in the motor in this situation.

4.3.2 Motor start-up sequence

If the one of the breakers is initially set to open, a start-up sequence may be simulated via a parameter event. You should do the following actions:

- Open the “Simulation Events” dialogue via the **Edit Simulation Events** button.
- Create a new parameter event (*.EvtParam*), configure the target element as being the “VSD Controller” common model, set the parameter to be “start_command” and set the value to 1.
- Configure the time instant of triggering the event exactly when the starting of the motor should occur.

The starting sequence is set via the of the characteristic “StartingCharacteristic” array stored inside the “VSD Controller” common model as an *IntMat* object.

An example of a start-up behaviour is shown in the figures below (refer to study case “SC05 - Motor starting (no re-acc) -RMS” of the previously mentioned FAQ example).

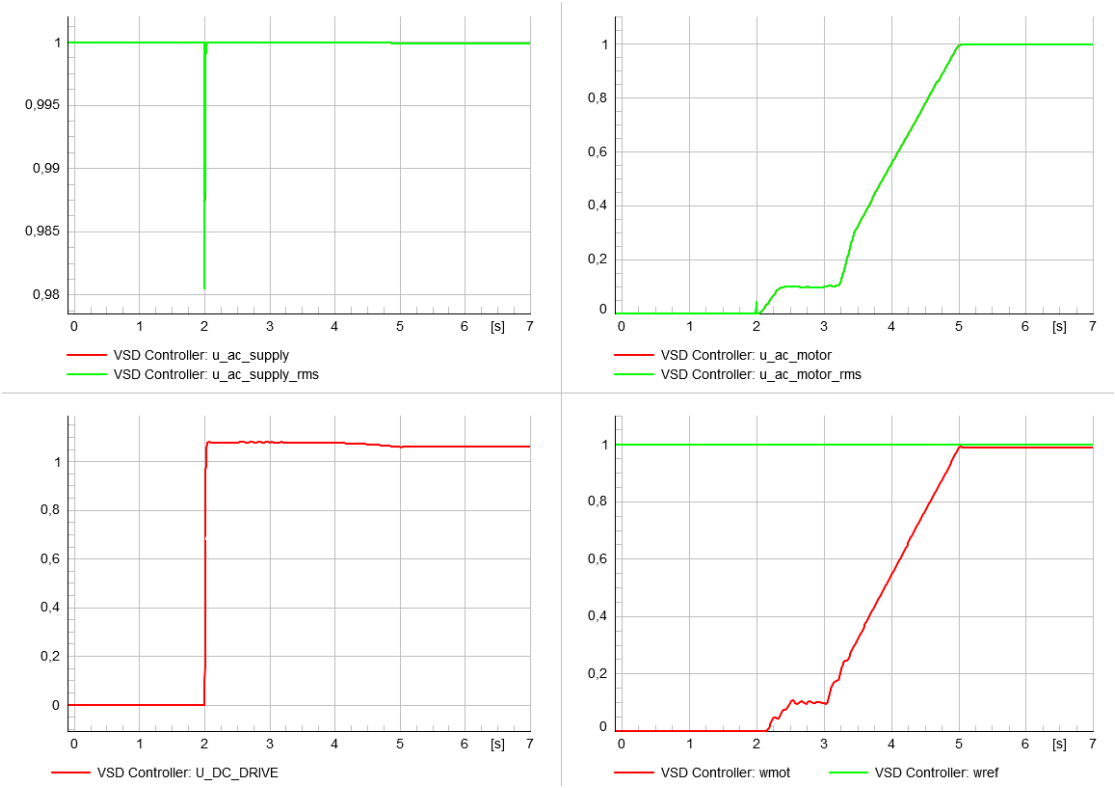


Figure 4.1: Example of a start-up procedure - electrical quantities

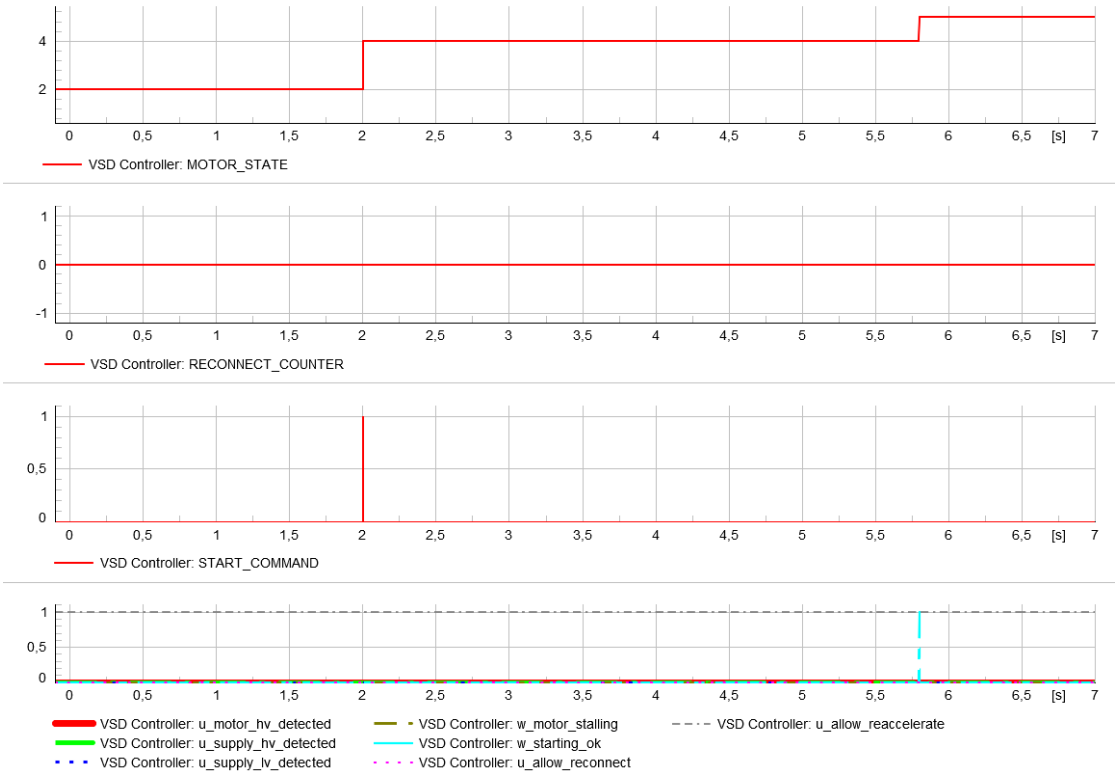


Figure 4.2: Example of a start-up procedure - state machine and logic signals

4.3.3 Steady state speed reference changes

To change the operating speed of the motor in time, the input signal *wref* of the block "VSD Controller" is available. This input signal may be changed during the simulation either via a parameter event or by adding a speed reference block to the VSD composite frame. Refer to the provided FAQ Example for a demonstration.

4.3.4 Motor reconnection (manual)

Whenever the VSD enters into fault mode, and provided that the parameter "SUPPORT_REACCELERATION" of the model *VSD Controller* is set to 0, the VSD may allow manual reconnection whenever the supply side voltage re-enters the normal operation band and the motor has reached a speed close to standstill (defined by $\omega_{mot} < \mathrm{WREF_MIN}$). In such a situation, the VSD will transition from the FAULT MODE state into the STANDSTILL state. A subsequent parameter event on the variable "start_command" of model "VSD Controller" is necessary (setting it to 1) in order to trigger the VSD to re-start the motor. The parameter event can be applied at any time, even prior the VSD entering the STANDSTILL state (it is assimilated to a switch with latch-on). If the parameter event is applied, then the VSD will re-trigger a start sequence and the motor will be re-started, as described in Section 4.3.2. Refer to the provided FAQ Example for a demonstration.

Note: If "SUPPORT_REACCELERATION"=0, then the VSD does not support automatic reacceleration as described in Section 4.3.5.

4.3.5 Motor re-acceleration (automatic)

The reacceleration option is activated by setting the parameter "SUPPORT_REACCELERATION" of the model *VSD Controller* to 1. If so configured, then the *VSD Controller* will try to reaccelerate the motor as soon as the post-fault voltage recovers within a certain safe operation band. The logic behind this is defined within the VSD state machine, decribed in the previous sections.

Refer to the study case "SC07 - Motor starting (re-acc) - RMS" of the FAQ example for a demonstration. Within Figure 4.3 the VSD can successfully ride through three consecutive faults and re-accelerate the motor.

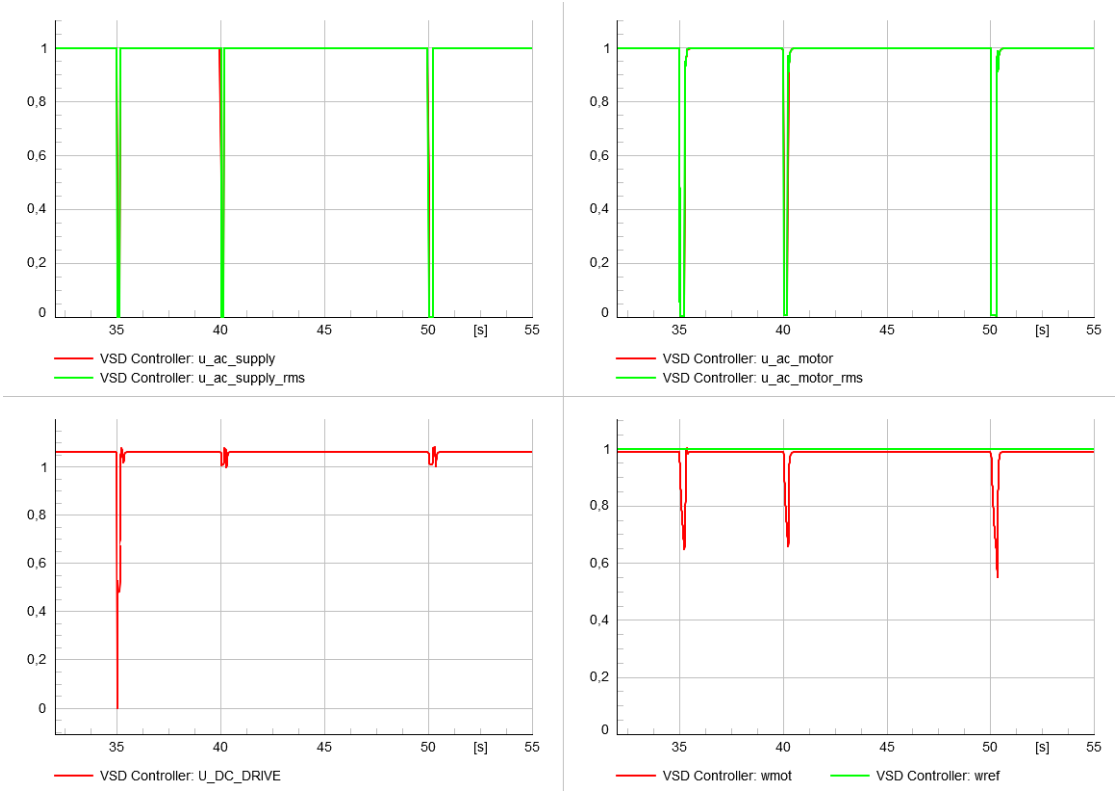


Figure 4.3: Example of VSD ride through with motor re-acceleration upon three consecutive voltage dips

Note: If “SUPPORT_REACCELERATION”=1, then the VSD does not support manual reconnection as described in Section 4.3.4.

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